

A Filtering Physics-Informed Neural Operator for Online Hydraulic Transient Teconstruction

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Abstract

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A Filtering Physics-Informed Neural Operator for Online Hydraulic Transient Reconstruction

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Key Points:

- A novel neural operator-based method for near real-time transient pressure reconstruction
- An information filtering mechanism to improve model performance
- A hybrid loss integrating real data, simulated data, and physics
- Fast and accurate transient reconstructions in hydraulic systems with sparse data

Abstract

Hydraulic transients pose significant risks to pipeline systems, including potential leaks and bursts. While traditional numerical methods are widely used for transient analysis, their reliance on detailed initial/boundary conditions limits their applicability in real-world scenarios. To overcome these challenges, this paper proposes a novel Filtering Physics-Informed Neural Operator (FPINO) framework for accurate and near real-time reconstruction of hydraulic transients from sparse observations. FPINO extends recent developments in Physics-Informed Neural Operators (PINOs), a class of machine learning models that incorporate physical laws directly into operator learning, to transient hydraulic applications. The FPINO framework introduces a time-weighted information filtering strategy that prioritizes the relevant observations, thereby accelerating convergence and improving prediction accuracy. Additionally, FPINO employs a hybrid loss function that integrates physics-based constraints with data-driven learning, ensuring both physical fidelity and adaptability to complex system behaviors. Numerical and experimental results show that FPINO outperforms the baseline PINO approach in reconstructing diverse transient scenarios. These results highlight the potential of FPINO as a robust and practical tool for real-time monitoring and analysis of hydraulic transients in pipeline systems.

Keywords: neural operator; partial differential equations; pipeline system; real-time reconstruction;

1. Introduction

Hydraulic transients, commonly referred to as water hammer, are characterized by rapid pressure fluctuations in pipelines that can result in significant physical damage, such as leaks (Hu et al., 2021) and bursts (Romano et al., 2014), to pipelines. To mitigate these risks and maintain the integrity of pipeline networks, various strategies have been proposed in the literature, such as optimal network design (Ulusoy et al., 2022), preventive valve control (Brentan et al., 2021), and smart water management (Owen, 2023). Transient analysis has emerged as an essential component of pipeline design and operation. By modeling and understanding the dynamics of pressure surges and flow instabilities, engineers can implement measures to prevent or control transient-induced damage, ensuring pipeline systems are both safe and efficient (Boulos et al., 2005).

Traditional model-based transient analysis (Chaudhry, 2014; Hwang & Chung, 2002; Zecchin et al., 2009; Zeng et al., 2022) has been the most common methods for decades. However, these methods typically require detailed knowledge of the system's physical properties and initial/boundary conditions (ICs/BCs). The accuracy of the numerical results is significantly affected by various uncertainties that are inevitable in real-world applications. On the other hand, the numerical computations can be computationally intensive, especially for large-scale pipe systems, further limiting their real-time analysis.

In recent years, the rapid development of online monitoring systems has offered a promising avenue for real-time analysis (Mounce et al., 2010; Lambrou et al., 2014; Kamaruidzaman & Nazahiyah, 2020). Researchers are increasingly focusing on data-driven techniques in water distribution systems for tasks, such as state estimation (Hajgató et al., 2021; Xing & Sela, 2022), leakage/burst detection (Romano et al., 2014; Wu & Liu, 2017), water quality prediction (Sokolova et al., 2022; Z. Li et al., 2024), and demand forecasting (Bata et al., 2020; Salloom et

al., 2021). These methods, particularly machine learning, utilize data to capture the underlying relationships in systems without explicit physical equations. This is particularly advantageous when the physical laws are not fully understood or uncertainties exist in the system parameters (Hutton et al., 2014). The integration of real-time data enables continuous learning and adaptation, making these methods more suitable for applications requiring real-time monitoring and control.

However, while a number of data-driven methods have been applied in steady-state flow, their application in the field of hydraulic transient remains relatively underexplored. The challenges of implementing data-driven methods in transient flow can be multifold. First, hydraulic transients often involve rapid flow/pressure changes over a very short time. Modeling the transient responses accurately requires capturing highly nonlinear and time-dependent behaviors, which can be challenging for data-driven methods that are typically designed for steady-state or slowly varying systems. Moreover, the collected data in network systems is often scarce due to the cost and logistical challenges of deploying and maintaining high-resolution sensors, hindering the training and validation of data-driven models for transient analysis.

Despite those challenges, a few efforts have been made in recent years for leak/burst detection and transient flow reconstruction. Bohorquez et al. (2021) introduce a two-stage approach for burst detection and identification in water pipelines by merging fluid transient waves with artificial neural networks (ANNs). The ANN-based technique demonstrated high accuracy in predicting burst locations. Ayati et al. (2022) presented a machine learning–assisted leak detection model that enhances the accuracy and reliability of identifying leaks in water distribution networks. In those two studies, the training datasets were generated using hydraulic simulation models since only a limited number of recorded leak or burst events are available in

real systems. To address the issue of data scarcity in transient reconstruction, Ye et al. (2022) developed a physics-informed neural network (PINN) (Raissi et al., 2019) method that integrates observed data with the physical laws governing hydraulic transients during the training process. The PINN can leverage the strengths of both data-driven methods and traditional physical models, thereby enabling accurate prediction of transient states even with sparse observations. The PINN-based method for a single pipeline was further extended to pipe networks (Ye et al., 2024) by integrating an efficient elastic water column (EWC) model (Zeng et al., 2022) into the neural network. The EWC model represented by a set of ordinary differential equations reduces the dimension of the input from the space-time domain to the time domain only. This characteristic not only reduces the size of the training dataset but also simplifies the prediction task from infinite space to several discrete nodes, making it easier to reconstruct the transient states over the whole network from the very sparse observation points. The developed PINN-based methods in single pipeline and network systems were both robust against various uncertainties in data and physical models.

While the PINN-based approaches show promise in transient flow reconstruction, they still face challenges and limitations in real applications. One significant gap is that the PINN model can only be used for a specific transient scenario once trained. This means a newly trained PINN model is required for a different transient scenario. However, the computational cost of training PINN models for each scenario can be prohibitive, particularly for large-scale systems or real-time applications where rapid decision and management are crucial.

Neural operators (Kovachki et al., 2023), a class of neural networks that do not hold that limitation (Kontolati et al., 2024), have recently attracted increasing interest in various engineering fields (Goswami et al., 2024; Lehmann et al., 2024; Li et al., 2024). The most

commonly used neural operators include the Fourier Neural Operator (FNO) (Li et al., 2021), Deep Operator Network (DeepONet) (Lu et al., 2021), and Graph Neural Operators (Li et al., 2020). The design of neural operators aims to approximate solutions to partial differential equations (PDEs) by learning mappings between function spaces. Regarding solving PDEs, the main difference between neural operators and the PINN is that neural operators can take ICs/BCs or observations into the input rather than the loss function of the neural network. This feature allows neural operators to learn the PDE solution from multiple scenarios while only one specific scenario can be considered in the PINN.

Recently, Gong et al. (2024) developed a surrogate transient model for gas pipelines based on FNO. The proposed model can serve as a solver for transient natural gas flow, efficiently providing accurate simulation results. However, the training process requires a complete dataset that includes the physical variables, such as the pressure and mass flow, in the entire space-time domain. Consequently, the trained neural operator relies heavily on numerical data, limiting its applicability in real-world scenarios where data is often highly sparse and various uncertainties associated with system parameters, process models, and measurement noise are involved. Furthermore, the trained model is mesh-based and specifically designed for forward problems with fully known ICs/BCs, which can often be addressed directly using traditional numerical methods. This restricts its capability to solve more complex inverse problems with unknown ICs/BCs.

This work aims to develop a neural operator-based framework for near real-time (within seconds) reconstruction of hydraulic transients without retraining. To address the challenges of unknown ICs/BCs and limited sensor data, DeepONet is adopted as the foundational architecture due to its flexibility and ability to handle irregular inputs. However, the vanilla DeepONet

struggles with temporally irrelevant input data, which hinders convergence and accuracy. To overcome this, a novel Filtering Physics-Informed Neural Operator (FPINO) that incorporates a time-weighted information filtering strategy is proposed in this paper. This mechanism captures the causal relationship between observations and predicted variables, enabling the model to prioritize relevant sensor data and significantly improve convergence ability. Additionally, the proposed FPINO employs a hybrid loss function that fuses data-driven and physics-based components, enabling robust reconstruction even under sparse measurements and system uncertainties. The key contributions of this work were summarized as follows:

- (1) Development of a novel FPINO method for near real-time transient reconstruction in pipeline systems;
- (2) Integration of a time-weighted information filtering mechanism to enhance model convergence by prioritizing causally relevant observations.;
- (3) Implementation of a hybrid loss function that combines simulated data, real measurements, and physical constraints for improved accuracy and robustness under uncertainties.

2. Materials and Methods

This section illustrates an improved physics-informed neural operator method for real-time and accurate transient reconstruction.

2.1 Mathematical representation of the transient flow

Assuming the flow is one-dimensional, slightly compressible, and occurs in an elastic pressurized pipe, the governing equations for transient flow can be expressed as follows (Wylie et al., 1993):

$$F_1 = A \frac{\partial q}{\partial t} + q \frac{\partial q}{\partial x} + g A^2 \frac{\partial h}{\partial x} + f \frac{|q|q}{2D} = 0 \quad (1)$$

$$F_2 = A \frac{\partial h}{\partial t} + q \frac{\partial h}{\partial x} + \frac{a^2}{g} \frac{\partial q}{\partial x} = 0 \quad (2)$$

where F_1 and F_2 represents the right side of the momentum equation and continuity equation, respectively; x is the distance along the pipeline; h is the piezometric head; q is the flowrate; t is the time since the beginning of the transient; a is the pressure wave speed; g is the gravitational acceleration; D is the internal pipe diameter; and $A = \pi D^2/4$ is the cross-section area of the pipeline, f is the Darcy–Weisbach friction factor.

The above partial differential equations can be solved by numerical transient models with known ICs/BCs. In this paper, the well-known method of characteristics (MOC) described in (Chaudhry, 2014) was employed to generate numerical datasets.

In real pipeline systems, transient pressure waves are typically subject to noticeable damping effects, which arise from several factors including unsteady friction (Brunone et al., 2000), pipe-wall viscoelasticity (Covas et al., 2005), and fluid-structure interaction (Tijsseling, 1996). This work considers the damping effects using the Kelvin-Voigt viscoelastic model. To account for the dynamic effects of pipe wall, a retarded viscoelastic term is introduced into the continuity equation (Eq. (2)), resulting in:

$$A \frac{\partial h}{\partial t} + q \frac{\partial h}{\partial x} + \frac{a^2}{g} \frac{\partial q}{\partial x} + \frac{2a^2 A}{g} \frac{\partial \varepsilon_r}{\partial t} = 0 \quad (3)$$

where ε_r denotes the retarded strain in the pipe wall. The viscoelastic response is characterized by a creep compliance function, defined as:

$$J(t) = J_0 + \sum_{i=1}^N J_i (1 - e^{-t/\tau_i}) \quad (4)$$

Here, $J_i = 1/E_i$ represents the creep compliance of the i -th Kelvin-Voigt element. Each element is described by an elastic modulus E_i and a retardation time $\tau_i = \mu_i/E_i$ with μ_i the viscosity of the corresponding dashpot.

Using this creep function, the total retarded strain $\varepsilon_r(x, t)$ is expressed as:

$$\begin{aligned} \varepsilon_r(x, t) &= \sum_{i=1}^N \varepsilon_{ri}(x, t) \\ &= \sum_{i=1}^N \left\{ \frac{\alpha D \rho g}{2e} \int_0^t [H(x, t - t') - H_0(x)] \frac{J_i}{\tau_i} e^{-t'/\tau_i} dt' \right\} \end{aligned} \quad (5)$$

where α is the pipe wall constraint coefficient; e is the wall thickness. This formulation enables the modeling of viscoelastic damping by accounting for the time-dependent deformation of the pipe wall. Further details on the numerical implementation of Eqs. (3)-(5) can be found in (Covas et al., 2005).

2.2 Inverse problem definition of online transient flow reconstruction

Conventionally, the transient states $\boldsymbol{\varphi}$ (i.e., pressure head h and flow q) in a pipe system are simulated by hydraulic transient models $\mathcal{M}$ (e.g., MOC) based on known ICs/BCs $\mathbf{b}$. This is referred to as a forward problem, represented by:

$$\mathcal{M}(\mathbf{b}) = \boldsymbol{\varphi} \quad (6)$$

However, the solution of the forward problem relies on completely known boundaries. In the case of a pipeline system with unknown ICs/BCs (see Fig. 1) but observed data $\mathbf{u}$, solving the transient problem becomes an inverse problem. These unknown ICs/BCs can arise from tank fluctuations, valve closures, pump failures, users' demand changes, or the need to analyze a specific pipe in a complex pipe network system. In inverse problems, instead of predicting the transient states constrained by ICs/BCs, our goal is to infer the spatiotemporal flow and pressure distribution along the pipeline from limited sensor data in near real time. The inverse problem is expressed by:

$$\mathcal{M}^{-1}(\mathbf{u}) = \boldsymbol{\varphi} \quad (7)$$

where $\mathcal{M}^{-1}$ is an inverse model that maps the observed data $\mathbf{u}$ back to the complete transient scenario.

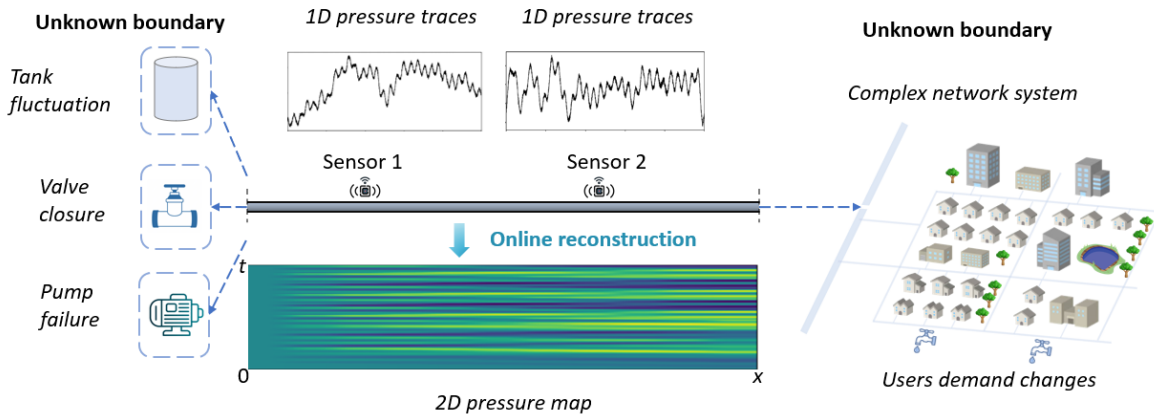


Fig. 1 Transient reconstruction in a pipeline system with unknown boundaries and two sensors.

In most real-world scenarios, $\mathcal{M}^{-1}$ is not explicitly known and must be approximated using optimization algorithms or machine learning models. Accurately solving the inverse problem is significantly more challenging than solving the forward problem for the following reasons: (1)

Unknown ICs/BCs, which are unmonitored, can significantly affect the transient behavior of the system. (2) The solution to the inverse problem is often non-unique and highly sensitive to measurement errors, requiring robust algorithms to ensure stability and accuracy. (3) The sparse placement of sensors provides incomplete spatial coverage, making it difficult to fully capture the transient states of the system. (4) Real-world uncertainties, such as deteriorated pipe conditions, uncharacterized wave damping, and measurement noise, further complicate the reconstruction process. (5) Conventional inverse transient analysis is designed for a specific scenario and requires repeated optimization or training for new transient events, making it unsuitable for online applications.

2.3 Physics-informed Neural Operator

Purely data-driven models rely on large amounts of labeled data to capture underlying patterns, whereas PINNs can achieve accurate predictions with significantly less data by leveraging physical constraints. Motivated by the success of PINNs, PINOs incorporate physical knowledge into the neural operator, leveraging both the flexibility and capability of operator learning and the robustness and interpretability of physics-informed models.

There are two common PINOs: the Physics-Informed FNO and Physics-Informed DeepONet. Compared to FNO, which requires well-defined inputs and outputs on a discretized grid, DeepONet offers a more flexible framework that can handle sparse and incomplete observations (Lu et al., 2022). Therefore, this study adopts DeepONet as the neural operator for the inverse problem.

2.3.1 Physics-informed DeepONet

The input of standard DeepONet consists of two components: the input function $\mathbf{u}$, which may include ICs/BCs, system parameters, and observation data; and the spatiotemporal coordinates $\mathbf{y} = (x, t)$, where x represents the spatial location and t the time, corresponding to any point in the solution domain.. The output is the physical variables $\boldsymbol{\varphi}(\mathbf{u}, \mathbf{y})$ at the point (x, t) . This unique structure allows DeepONet to handle sparse and incomplete data more effectively, making it particularly flexible and advantageous for the transient reconstruction in pipe systems.

Correspondingly, the DeepONet consists of two neural networks that work together to represent the operator: branch network and trunk network (Lu et al., 2021). The branch network is used to process the input function $\mathbf{u}$. It takes the discretized input function (a finite set of values representing the function) and outputs a latent representation. The trunk network processes the spatiotemporal coordinates $\mathbf{y}$ at which the output function is evaluated and also produces a latent representation. The latent representations from the branch and trunk networks are then combined using a dot product to formulate the final output $\boldsymbol{\varphi}(\mathbf{u}, \mathbf{y})$, which are the physical variables at the spatiotemporal point $\mathbf{y}$ under dynamic conditions $\mathbf{u}$.

The DeepONet is trained by minimizing a loss function that measures the discrepancy between the predicted and true output functions. In practice, however, the sparsity, low resolution, and incompleteness of field data limit its ability to approximate the ground-truth physical process. Physics-informed DeepONet (Wang et al., 2021) overcomes the shortcoming by formulating a hybrid loss function that combines training data with physical laws (often exhibit as PDEs). By doing so, the physics-informed DeepONet effectively regularizes the learning process, guiding the network towards physically plausible solutions even when the available data is sparse or noisy.

2.3.2 Drawback of villina DeepONet

The output of DeepONet forward propagation consists of the physical states $\boldsymbol{\varphi}_s$ at a specific sampling point $\mathbf{y}_s = (x_s, t_s)$. However, the input of the branch network is composed of BCs or observations over the entire analyzed period. In many cases, the states $\boldsymbol{\varphi}_s$ to be determined depend only on the initial conditions and BCs/observations within the interval $[0, t_s \pm \Delta t]$. This interval is determined by the physics of the problem, such as the wave speed and spatial relationships. Any input data outside this interval are physically irrelevant and considered irrelevant information.

In pipeline systems, a preliminary analysis based on an MOC grid is performed to determine the essential information required to solve the condition at a specific sample location. For the forward problem with known ICs/BCs (as illustrated Fig. 2(a)), the flow and head at the sample can be directly determined using a forward model (e.g. MOC) only using ICs and the BCs within the time intervals $[0, t_{b,1}]$ for the left boundary and $[0, t_{b,2}]$ for the right boundary. The time $t_{b,i}$ are calculated as $t_s - d_i/a$, where d_i is the distance between the sample and the corresponding boundary. The conditions required for solving the sample are referred to as essential information. Conversely, any boundary conditions outside the time intervals $[0, t_{b,1}]$ and $[0, t_{b,2}]$ are not required for the solution and are therefore classified as irrelevant information.

For the inverse problem with unknown boundaries but available sensors (see Fig. 2(b)), the extended Method of Characteristics (EMOC) (Ye et al., 2025), an inverse modelling method for transient analysis, is employed to determine the essential information required to solve a sample. There are two possible scenarios in the inverse problem. (1) The sample is located between the sensors (Fig. 2(b)). In this case, the analysis for the sample remains the same as in the forward problem by regarding the sensor data as “boundary conditions”, while the true ICs/BCs are not

directly required in this scenario. Note that the ICs are assumed to be in steady state and can be computed by the measured heads at $t = 0$ s. (2) The sample is located between a sensor and an unknown boundary (Fig. 2(c)). Based on the EMOC framework, the essential information only includes the sensor data with the intervals $[0, t_{obs,1}]$ for the left sensor and $[0, t_{obs,2}]$ for the right sensor. The time $t_{obs,2} = t_s + d_2/a$ where d_2 is the distance between Sensor 2 and the right boundary. The time $t_{obs,1} = t_s - (d_1 - d_2)/a$ where d_1 is the distance between the sensors. The remaining sensor data are irrelevant.

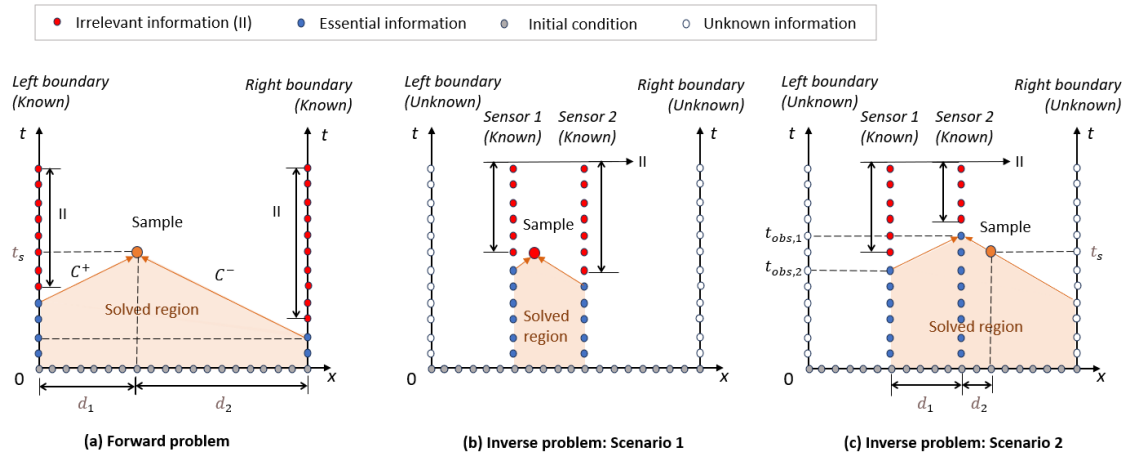


Fig. 2 Preliminary analysis to solve the states of samples (a) Forward problem; (b) Inverse problem: Scenario 1; (c) Inverse problem: Scenario 2.

Including irrelevant information can weaken the true causal relationships between input observations and output predictions. This increases the burden on the optimizer to distinguish relevant inputs, thereby slowing convergence. Furthermore, feeding in irrelevant data increases the risk of overfitting, where the model memorizes patterns that do not reflect the underlying physics but instead arise from irrelevant observations.

2.4 Filtering physics-informed neural operator (FPINO)

2.4.1 The workflow of FPINO-based transient reconstruction

This paper proposes an FPINO framework to achieve near real-time transient reconstruction. Fig. 3 illustrates the workflow of the FPINO framework, which consists of three stages: data preparation, neural network training, and online prediction. The FPINO incorporates a time-weighted information filtering strategy to prioritize relevant data, ensuring that the reconstruction process for each sample focuses on the most impactful information. In addition, a hybrid physics-driven and data-driven loss function that integrates physical constraints (governing equations), simulation data, and observed data is formulated to enable robust and efficient transient flow reconstruction even with sparse observations and various uncertainties. The final architecture of the proposed FPINO is illustrated in Fig. 4.

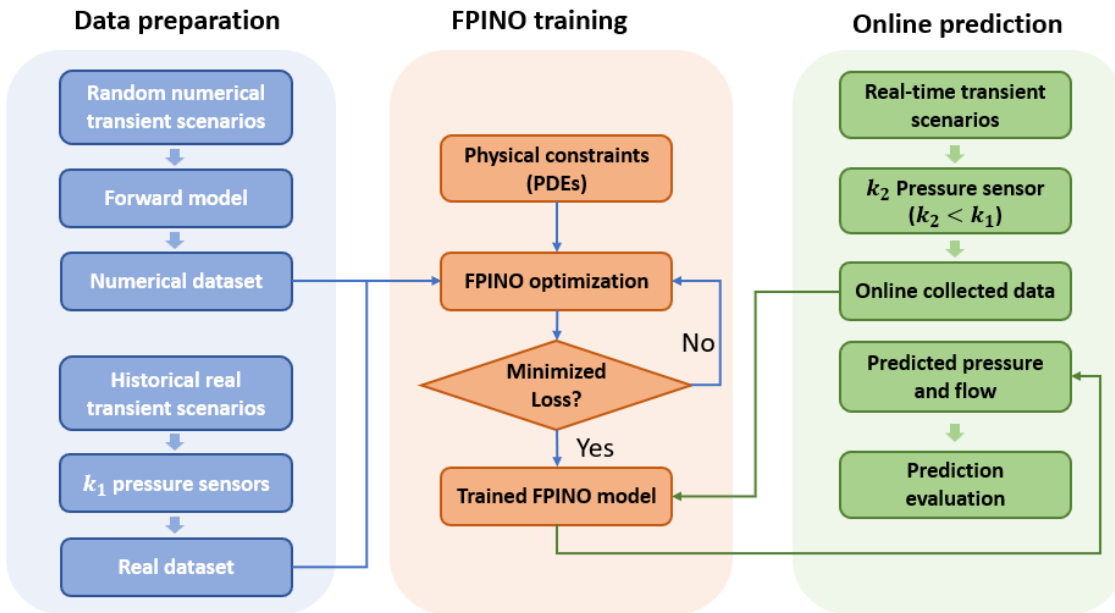


Fig. 3 The workflow of the FPINO.

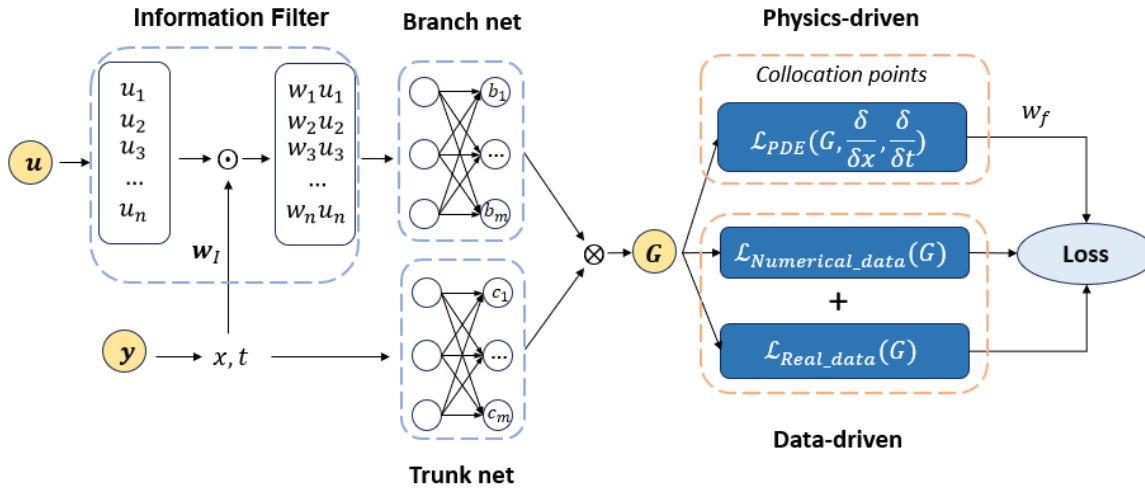


Fig. 4 The architecture of the FPINO with a hybrid loss.

2.4.2 Information filter

Before feeding the observation data into the branch network of DeepONet, an information filter is applied to assign weights to each observation point based on its relevance to the prediction point. This is achieved using a differentiable stiff sigmoid function:

$$w_I = \text{sigmoid}(-k * (t_{obs} - t_s - \tau)) \quad (8)$$

where t_{obs} is the time at the observation point, and t_s is the time at the sample (prediction point). The constant k controls the steepness of the sigmoid curve. A stiff sigmoid (with a large k) creates a sharp transition between the relevant and irrelevant regions, effectively filtering out irrelevant information while preserving the essential data for accurate prediction. The constant τ is the time offset, representing the arrival time of wave propagation from the observation point to the prediction point. For example, Sample 2 in Fig. 2(b) is located between a sensor and an unknown boundary. The corresponding τ is calculated by $\beta d_2/a$, where β is the constant to

account for wave damping and other uncertainties in the wave propagation. In this paper, β is set to be 1.2.

By applying the production of the weights and the observed data, the information filter assigns higher weights (near one) to the essential information and lower weights (near zero) to the irrelevant information. This filtering process helps the neural network focus on essential input information, thereby improving training efficiency and prediction accuracy.

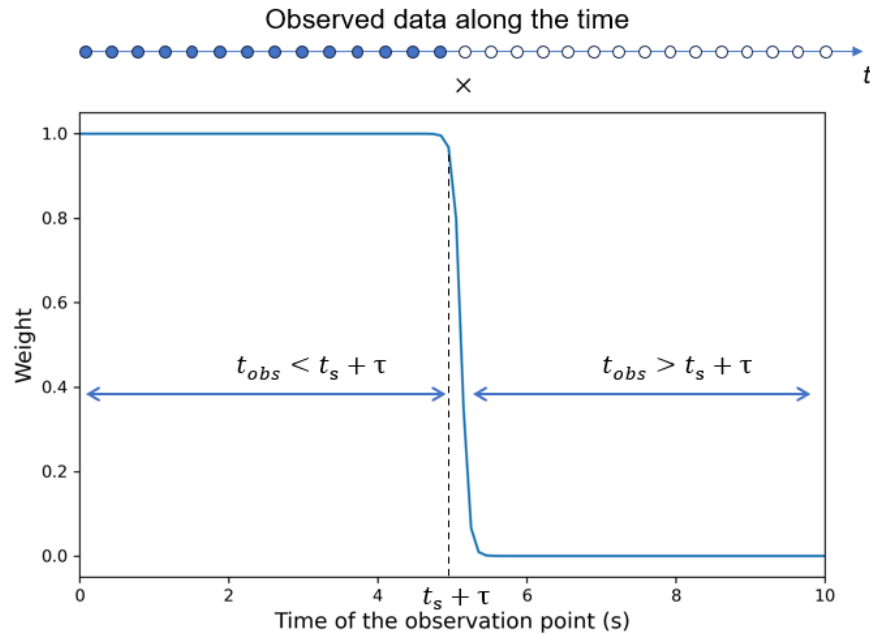


Fig. 5 An example of the weight function with $k = 20$, $t_s = 5$ s, and $\tau = 0.12$ s.

2.4.3 Hybrid loss function

The FPINO framework utilizes a hybrid loss function that integrates both physics-driven (governing PDEs) and data-driven (both simulated and real data) components to enhance the prediction accuracy of physical systems. The loss function is expressed as

$$L(\boldsymbol{\theta}) = w_f L_{PDE} + w_{sim} L_{simulated_data} + w_{real} L_{real_data} \quad (9)$$

The hybrid loss function is comprised of three losses: L_{PDE} , $L_{simulated_data}$, and L_{real_data} . The term L_{PDE} measures the violation of physical constraints. The term $L_{simulated_data}$ measures the discrepancy between the predicted and simulated data which can be obtained from a numerical solver. The losses L_{PDE} and $L_{simulated_data}$ are both model-based, forcing the operator to match the physical model. However, due to various uncertainties in reality, the physical model usually is unable to perfectly represent the real physical phenomenon. In water pipelines, for instance, no such model can completely characterize the dispersion and dissipation of the wave propagation (also known as wave damping). In this case, the term L_{real_data} that measures the discrepancy between the predicted and real data is integrated into the loss function to extract the hidden information from measured data. The weighting values w_f , w_{sim} and w_{real} are designed to balance those three losses L_{PDE} , $L_{simulated_data}$, and L_{real_data} .

The expressions of losses L_{PDE} , $L_{simulated_data}$, and L_{real_data} are given by:

$$L_{PDE} = \frac{1}{(N_{s1}+N_{s2}) \times N_f} \sum_{k=1}^{N_{s1}+N_{s2}} \sum_{j=1}^{N_f} (\hat{F}_1^2(x_f^j, t_f^j; \boldsymbol{\theta}) + \hat{F}_2^2(x_f^j, t_f^j; \boldsymbol{\theta})) \quad (10)$$

$$L_{simulated_data} = \frac{1}{N_{s1} \times N_{sd}} \sum_{k=1}^{N_{s1}} \sum_{j=1}^{N_{sd}} (\hat{\boldsymbol{\varphi}}_{sd}(x_{sd}^j, t_{sd}^j; \boldsymbol{\theta}) - \boldsymbol{\varphi}_{nd}(x_{sd}^j, t_{sd}^j))^2 \quad (11)$$

$$L_{real_data} = \frac{1}{N_{s2} \times N_{rd}} \sum_{k=1}^{N_{s2}} \sum_{j=1}^{N_{rd}} (\hat{\boldsymbol{\varphi}}_{rd}(x_{rd}^j, t_{rd}^j; \boldsymbol{\theta}) - \boldsymbol{\varphi}_{rd}(x_{rd}^j, t_{rd}^j))^2 \quad (12)$$

where $\boldsymbol{\theta}$ is the weights and bias of the neural network. N_{s1} and N_{s2} are the number of numerical scenarios and real scenarios, respectively. N_f is the number of collocation points (x_f, t_f) where the governing equations are enforced during the training process. The left side of the governing

equations (Equations (1) and (2)) $\hat{F}_1$ and $\hat{F}_2$ are evaluated at these collocation points by using the predicted physical variables. The hat represents predicted values by the neural network. The derivatives in $\hat{F}_1$ and $\hat{F}_2$ are obtained through the automatic differentiation library of neural networks (autograd). It is noted that no observed data are required for the PDE loss L_{PDE} . Thus, the collocation points can be at any spatiotemporal point for both numerical and real scenarios. In this study, the collocation points consist of all the grid points in the numerical and real scenarios. N_{sd} is the number of data points (x_{sd}, t_{sd}) in simulated scenarios, covering the entire solution domain. N_{rd} is the number of collected data points (x_{rd}, t_{rd}) in real data scenarios, which depends on the number of deployed sensors. $\boldsymbol{\varphi}$ is the vector of observed physical variables. While $\boldsymbol{\varphi}_{sd}$ consists of all the involved physical variables in the numerical scenarios, $\boldsymbol{\varphi}_{rd}$ in real scenarios depends on the type of used sensors and it is typically the pressure head in pipeline systems.

2.4.4 Formulation of training dataset

Based on the proposed hybrid loss function, the training data preparation for the transient reconstruction problem involves two datasets, the numerical dataset and the real dataset. The data preparation for the input and output of FPINO is illustrated in Fig. 6.

(1) Raw datasets

To generate the raw numerical dataset, a forward model, which is MOC in this study, is first built to simulate N_{s1} random numerical transient scenarios (e.g. random valve closures). The transient source is assumed to be outside or at the boundary of the analyzed pipeline. Each scenario contains $N_T \times N_X$ grid points in the x - t domain, where N_T is the number of time points and N_X is the number of space points along the pipeline. At each grid point, the coordinates x and t and the

corresponding physical variables (pressure head h and flow q) are bound to formulate a sample of the raw simulated dataset. Therefore, the total size of the raw numerical dataset is $N_{s1} \times N_T \times N_X \times 4$.

For the real dataset, N_{s2} historical transient scenarios are collected from N_{obs} pressure sensors (where N_{obs} denotes the total number of sensors). Each sample contains the coordinates and the corresponding pressure head. Consequently, the size of the real dataset is $N_{s2} \times N_T \times N_{obs} \times 3$.

(2) Input block

The raw datasets are divided into observation points and collocation points to enforce the observed data and physical constraints (PDEs), respectively.

Each observation sample in the input block $\mathbf{X}_1$ consists of the coordinates (x_{sd}, t_{sd}) and the observed pressure traces $\mathbf{h}_k$ at k sensor as input features. The observed pressure heads are flattened into one row with a length of $k \times N_T$. Thus, the size of the input vector is $(2 + k \times N_T)$ for each sample and the total size of the input block is $N_{s1} \times N_T \times N_X \times (2 + k \times N_T)$. Similarly, the input blocks $\mathbf{X}_2$ for real dataset has a total size of $N_{s2} \times N_T \times N_{obs} \times (2 + k \times N_T)$. It is worth noting that the pressure heads at only k ($k < N_{obs}$) sensor locations are used as input features for specifying the transient scenarios, while the pressure head data from all N_{obs} sensor locations are utilized to compute the data loss during the training stage. This enables the use of additional sensors to enrich the training dataset and improve model performance. However, once the FPINO model is trained, a reduced number of sensors (k) is sufficient to accurately reconstruct the transient scenario.

The collocation points of the numerical datasets share the same samples as in $\mathbf{X}_1$. However, these points do not require observed physical variables. Instead, they are used to enforce the governing physical constraints by evaluating the residuals of the equations at these points. For this reason, the collocation points in the real datasets are not restricted to the locations of available sensors and can be any point in the x - t domain. In this study, they are chosen as all the grid points. By bounding the collocation points in the simulated dataset and real dataset, the total size of the input block for collocation points is $(N_{s1} + N_{s2}) \times N_T \times N_X \times (2 + k \times N_T)$.

(3) Output block

The output block $\mathbf{Y}_1$ for the simulated dataset consists of both the pressure $h(x_{sd}, t_{sd})$ and flow $q(x_{sd}, t_{sd})$, while the output block $\mathbf{Y}_2$ for the real dataset consists of only the pressure $h(x_{rd}, t_{rd})$. Therefore, the total sizes of the output blocks are $N_{s1} \times N_T \times N_X \times 2$ for the simulated dataset and $N_{s2} \times N_T \times N_{obs} \times 1$ for the real dataset.

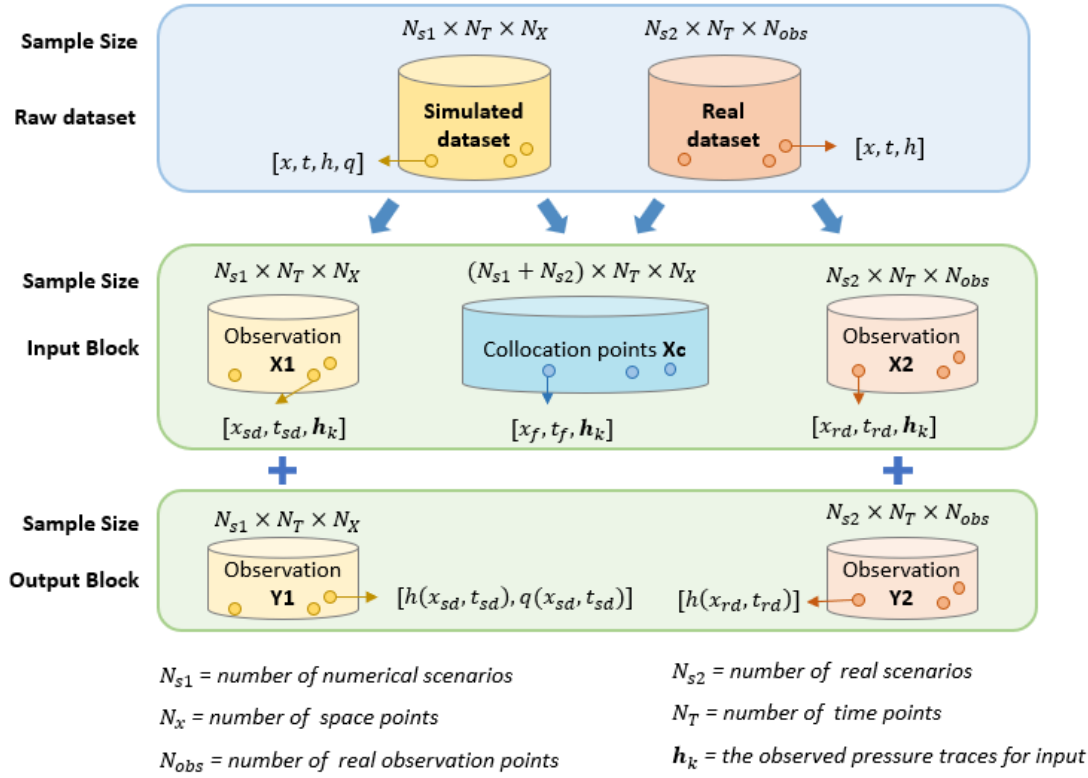


Fig. 6 Dataset generation flowchart. The size of each dataset is presented on the top of the dataset shaped by a cylinder.

2.4.5 Performance evaluation metrics

The root mean squared error (RMSE, also known as L2 error) and mean absolute error (MAE) are adopted to quantify the discrepancy between the predicted and actual values:

$$\text{L2 error} = \sqrt{\frac{\sum_{i=1}^N (\hat{\varphi}_i - \varphi_i)^2}{N}} \quad (13)$$

$$\text{MAE} = \frac{\sum_{i=1}^N |\hat{\varphi}_i - \varphi_i|}{N} \quad (14)$$

in which N is the number of data points evaluated, φ_i is the actual value of the transient state which could be either head h or flow q , and $\widehat{\varphi}_i$ is the corresponding predicted value.

3. Numerical Case Studies

3.1 System setup

A typical reservoir-pipe-valve (RPV) system shown in Fig. 7 was used to validate the proposed FPINO. The pipeline is made of high-density polyethylene (HDPE) with a length of 500 m. The water level of the reservoir is 50 m. The pipeline has a wave speed of 1000 m/s, a diameter of 0.2 m, and a steady-state friction factor of 0.012. The flow is initially in steady state. Three pressure sensors at $x = 280$ m, 360 m, and 480 m are deployed along the pipeline in the training stage, while only one pressure sensor at $x = 360$ m is required as input in the testing stage.

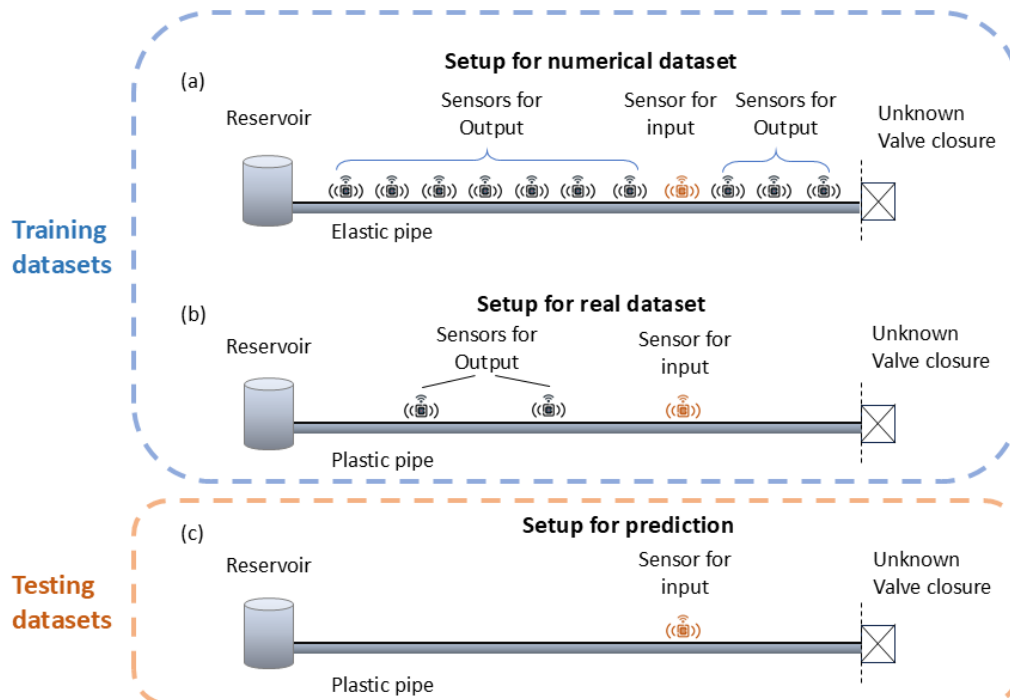


Fig. 7 The pipeline system with a reservoir, an end valve, a plastic pipe, and a pressure sensor. The initial flow and valve closure were assumed to be unknown.

3.2 Data generation

The transient events are generated with varied initial flow rates ranging from 0.02 m³/s to 0.05 m³/s and different valve closure patterns. The valve closure patterns include two-phase and three-phase linear closures, as illustrated in Fig. 8. The two-phase linear closure involves a single turning point where the valve's opening changes linearly in two segments. The three-phase linear closure involves two distinct turning points, creating three segments of linear closure. The total time for each valve closure event is sampled from 1 to 3.5 seconds. By randomly setting the initial flow rate and valve closure patterns, diverse transient scenarios can be generated to cover a wide range of transient behaviors in the RPV system.

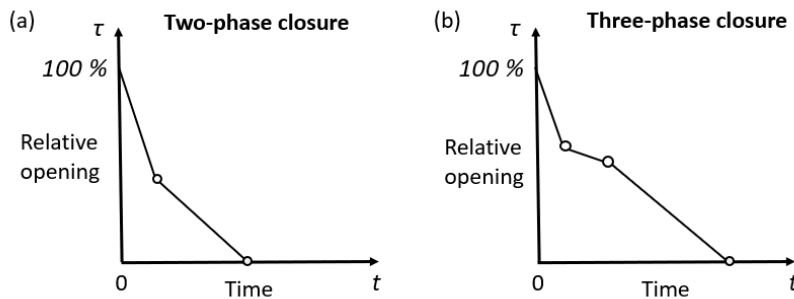


Fig. 8 Examples of (a) the two-phase closure and (b) the three-phase closure.

In the real world, the parameters relating to the viscoelasticity of the plastic pipe are difficult to obtain and calibrate. Therefore, the numerical dataset was generated using the standard MOC without knowing the viscoelastic properties of the plastic pipeline. A total of 500 transient scenarios were simulated, with 250 scenarios for each of the two closure types (two-phase and

three-phase closures). Each transient scenario is simulated for 10 seconds with a time step of 0.01 s, resulting in 1,000 time points in the time series and 51 spatial points along the pipeline. To reduce the dimensionality of input and the training workload, the simulated data is evenly downsampled in both time and space. Each scenario is reduced to 200 time points and 26 spatial points. Due to the sufficient available data from the simulation, observations are assumed to be located at all grid points of the pipeline, as shown in Fig. 7(a). Consequently, the observation dataset for the 500 simulated transient scenarios contains a total of 2,600,000 samples. The input of each sample includes its coordinates (x, t) and the 200 the pressure data points observed at a distance of 360 m from the reservoir, resulting in an input vector size of 202.

To generate the real dataset, the viscoelastic behavior of the pipe was incorporated into the transient simulation using the Kelvin-Voigt model. Then, the same 500 transient scenarios are simulated using the viscoelastic model for the real dataset generation. Fig. 9 shows a comparison of the observed pressure at 350 m from the simulated and real datasets under the same valve closure pattern. The real pressure trace simulated by the viscoelastic model shows obvious attenuation in the pressure peaks, better reflecting real-world conditions.

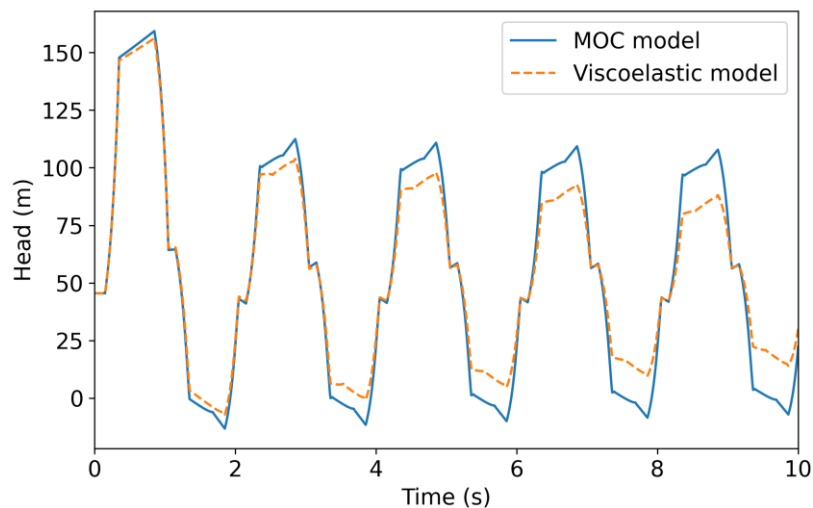


Fig. 9 Comparison of pressure traces simulated by the MOC model and the MOC integrated with the viscoelastic model.

Among the real transient scenarios, 400 transient scenarios were put in the training dataset with the sensor at $x = 360$ m providing the input data, and all three sensors provide the observed data for the output. The remaining 100 scenarios are used as testing data to evaluate the network performance. In total, the observation dataset includes 5,200,000 samples, with 2,600,000 samples from the simulated dataset and the other 2,600,000 samples from the real dataset. A summary of the simulated dataset and the real dataset is provided in Table 1.

Table 1 Summary of the simulated dataset and real dataset.

	Simulated dataset	Real dataset
Viscoelasticity	Not considered	Considered
Training scenarios	400	400
Testing scenarios	0	100
Observation samples	2,600,000	300,000
Collocation samples	2,600,000	2,600,000
Sensor location for input (m)	$x = 360$ m	$x = 360$ m
Sensor locations for output (m)	All internal points	$x = 280, 360, \text{ and } 480$ m

3.3 Hyperparameters

Hyperparameters of ANNs such as learning rate, network width and depth, govern the architecture, training process, and optimization of the network and are crucial for the performance and success of the ANN prediction. Choosing appropriate values for these

hyperparameters can significantly impact the ANN's ability to learn effectively, generalize to unseen data, and converge efficiently.

Grid search is a foundational tool for hyperparameter tuning where a predefined set of hyperparameters and their possible values are explored to find the combination that provides the best model performance on a validation dataset. It is one of the most straightforward yet effective methods for optimizing hyperparameters in machine learning and deep learning. The final FPINO hyperparameters listed in Table 2 were carefully selected through a combination of grid search and empirical testing to optimize model performance.

Table 2 Best values of hyperparameters

Hyperparameters	Observation points
Number of Epochs	100,000
Learning rate	0.001
Batch size	16
Activation function	tanh
Number of branch hidden layers	10
Number of trunk hidden layers	8
Width of the branch network	60
Width of the trunk network	30

3.4 PINO vs FPINO

Both PINO vs FPINO demonstrate high computational efficiency, requiring approximately 15 seconds to reconstruct 100 transient events, or about 0.15 seconds per event. The reconstruction accuracy of PINO and FPINO methods is evaluated using the L2 error and MAE metrics across different combinations of training scenarios. The combinations of simulated/real scenarios include 0/400, 50/350, 100/300, 200/200, 300/100, 350/50, and 400/0, maintaining a total of 400 training scenarios for each combination. The weights of w_f , w_{sim} and w_{real} are all set to be 1.

The L2 errors of other combinations are shown in Fig. 10, while the MAEs followed a similar trend and were consistently lower in FPINO. For reference, the MAE values over the testing samples of each case are summarized in Table A5.1 in the Appendix. For scenarios with only real data (i.e., the 0/400 case) or with only simulated data (i.e., the 400/0 case), both PINO and FPINO show higher variability in L2 errors. The results demonstrate that too much reliance on simulated data may lead to overfitting to the inaccurate system model, while an overemphasis on real-world data can amplify data sparsity issues. A balanced hybrid loss can effectively combine the strengths of simulated data for learning physical dynamics and real-world data for fine-tuning predictions, achieving robust and accurate model performance.

Across all scenario combinations, it is obvious that FPINO consistently achieves lower mean and median L2 errors compared to PINO, indicating that FPINO generally produces more accurate predictions. This is because the information filter mechanism in FPINO effectively reduces the impact of irrelevant or irrelevant data, focusing the model's learning on essential information.

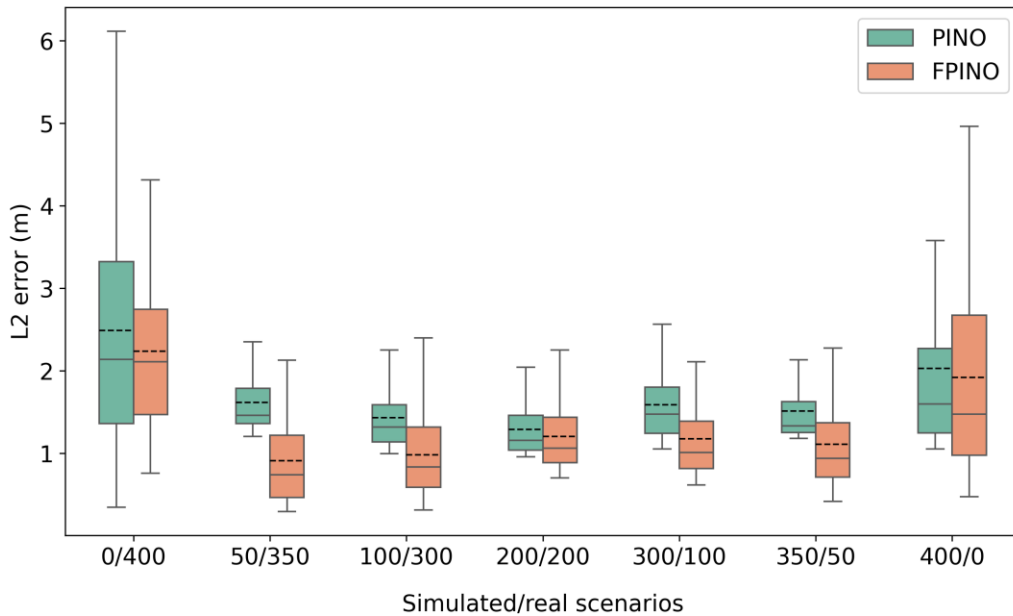


Fig. 10 Comparison of L2 errors between PINO and FPINO models across different combinations of simulated and real scenarios.

Fig. 11 compares the pressure predictions of PINO and FPINO with the ground truth at $x = 240\text{m}$ for four transient scenarios in the 350/50 case. The L2 errors for these four samples are [0.95, 1.27, 1.87, 2.41] m for PINO and [0.50, 0.67, 1.03, 2.05] m for FPINO, respectively. On average, FPINO achieves 57.7% higher accuracy compared to PINO. The analysis of these scenarios suggests FPINO closely matches the ground truth across all pressure peaks and troughs. At the same time, PINO shows slight deviations, particularly at higher pressure peaks, indicating its limitations in capturing detailed dynamics. Fig. 12 shows a comparison between the true values (left) and predicted values (right) by FPINO for both the pressure and flow across all the spatial and temporal grids. The patterns are mostly consistent with the true values. This emphasizes the robustness of the model in accurately capturing complex hydraulic transients.

Overall, FPINO outperforms PINO across all evaluated scenarios, especially in handling sharp transient dynamics, showcasing superior learning capabilities for transient events than the vanilla PINO.

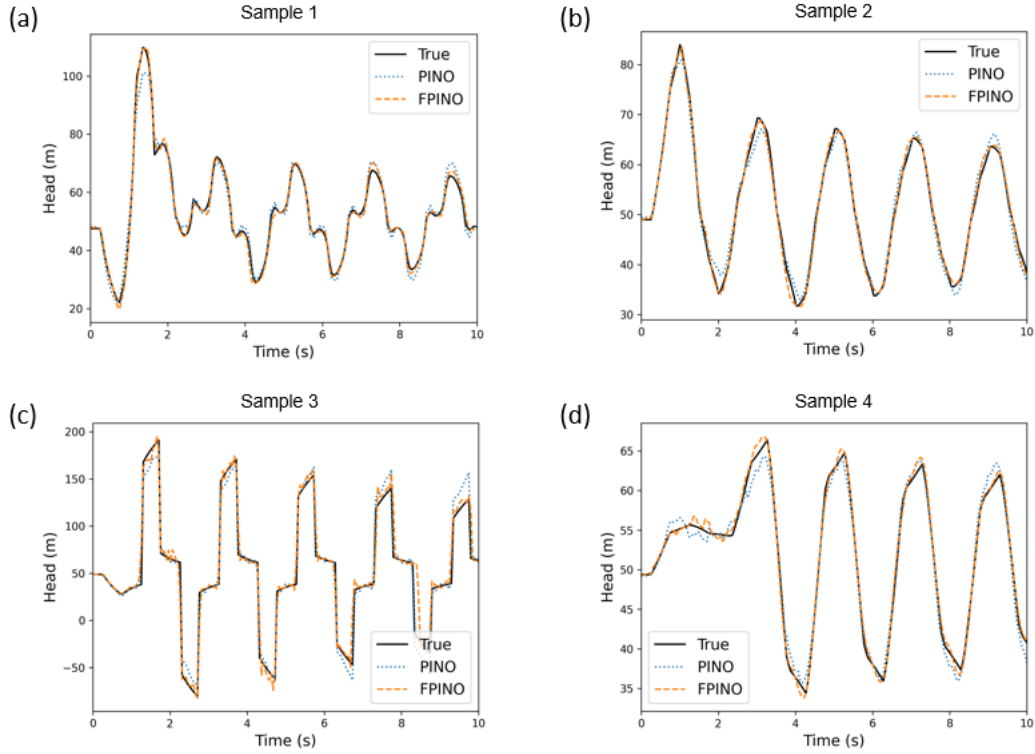


Fig. 11 Four samples of reconstructed pressures by PINO and FPINO in the 350/50 case.

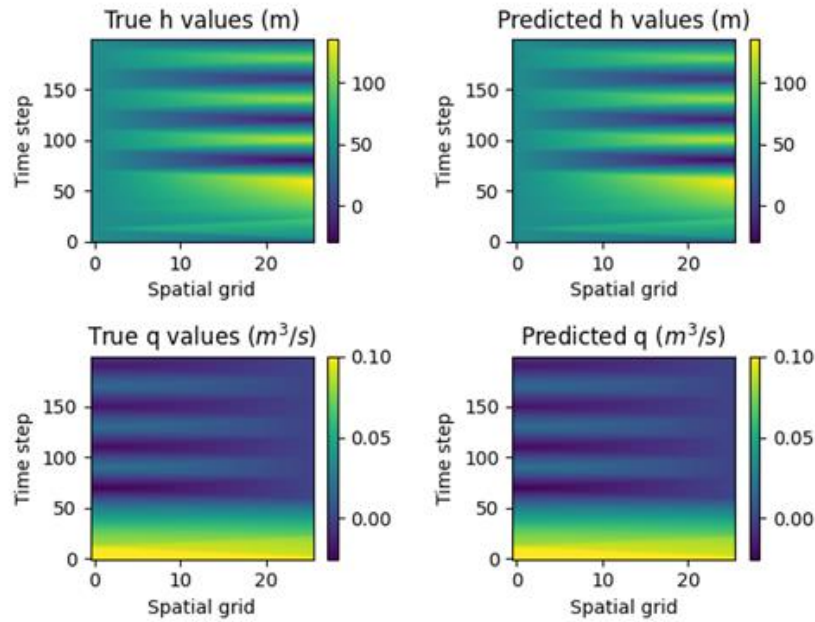


Fig. 12 Comparison between predicted physical variables by FPINO and the ground truth across the spatial and temporal grids: (a) head and (b) flow.

3.5 The impact of loss weights

In FPINO, the hybrid loss function integrates physics-driven and data-driven components. This balance ensures that the model adheres to physical laws while simultaneously capturing real-world observations. Due to different information and types of uncertainties from the two components, the loss weights w_f , w_{sim} , and w_{real} (see Equation (9)) play pivotal roles in determining their relative contributions during the training. Finding the optimal balance between these components can be critical to achieving accurate, stable, and generalizable predictions.

To investigate the effects of loss weights on prediction performance, the FPINO model in 200/200 case is evaluated by varying the values of w_f , w_{sim} , and w_{real} , independently, while keeping the other two weights fixed at 1. Each weight is chosen to be 0, 0.001, 0.1, 1, 10, 100, and 1000, respectively. The effects of varying weights are illustrated in Fig. 13, which shows the L2 errors and MAEs of FPINO for varying loss weights.

As seen from Fig. 13, the sensitivities of weights w_f , w_{sim} , and w_{real} vary significantly and exhibit different trends within the ranges $[0, 1]$ and $[1, 1000]$. The weight w_f (for PDE residuals) appears to be less sensitive compared to w_{sim} and w_{real} , as evident from the relatively small changes in errors across varying w_f values. This could be attributed to the fact that the physical information is also inherently embedded in simulated data. Consequently, the simulated data implicitly enforces physical constraints, reducing the model's reliance on the explicit physics-driven loss (L_{PDE}). This overlap makes the weight w_f less sensitive.

On the contrary, both the L2 error and MAE increase sharply as w_{sim} decreases from 1 to 0, indicating that simulated data is crucial for learning the underlying physics. However, when w_{sim} is too large ($w_{sim} \geq 10$), the model prioritizes physical knowledge over observed data,

leading to a slight increase in error. This demonstrates the potential drawbacks of over-relying on physics-based constraints, which may fail to account for real-world complexities. However, overemphasizing observed data (w_{real} increases from 1 to 1000) also leads to error growth. This occurs because the model becomes overly sensitive to the data sparsity, and inconsistencies inherent in real-world data. Overall, the results demonstrate that balancing the contributions of physical knowledge and data is critical for the accurate predictions of FPINO.

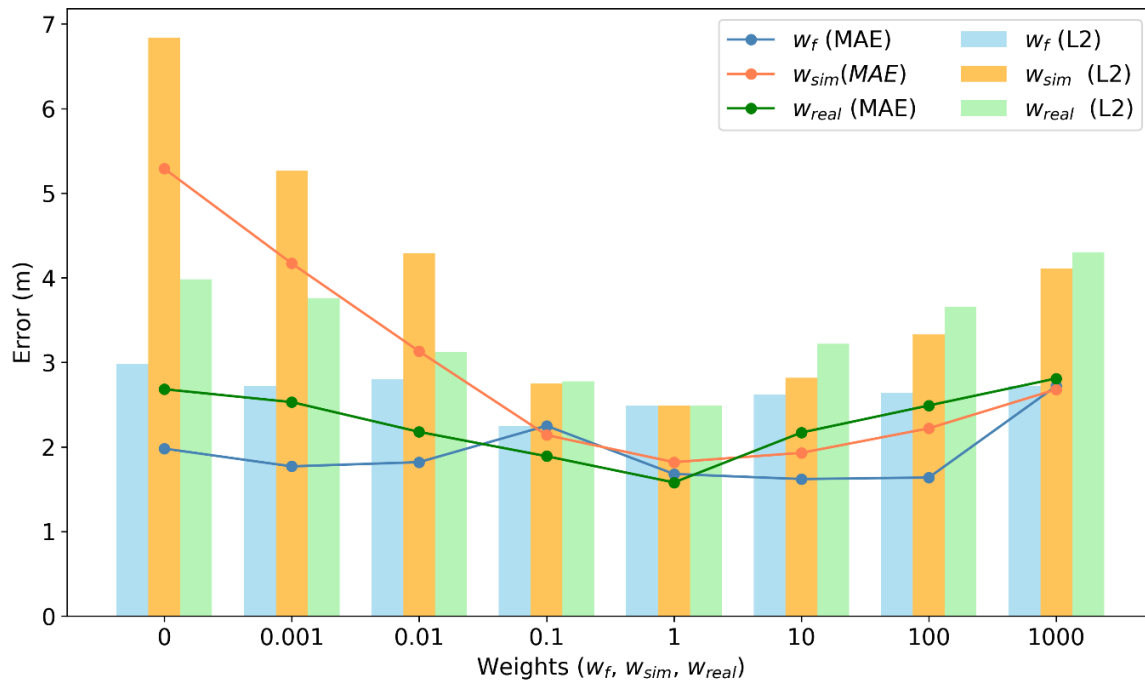


Fig. 13 The L2 errors and MAEs from FPINOs with varying weights by changing w_f , w_{sim} , or w_{real} , independently, while keeping the other two weights fixed at 1.

4. Experimental Case Study

The applicability of the proposed FPINO in reconstructing hydraulic transients was further validated on a laboratory pipeline located in the Robin Hydraulics Laboratory at the University

of Adelaide. The schematic of the laboratory experimental setup is shown in Fig. 14. This pipeline consists of a copper pipe with a total length of 37.21 m and a slight inclination. The inner diameter of the copper pipe is 0.02214 m. The wave speed was calculated using the theoretical formula as $a = 1319$ m/s. A pressure vessel on the right end of the pipe supplies pressurized water to the pipeline. An inlet gate valve connects the pipeline to the pressure vessel, regulating the flow of pressurized water into the system. The left end of the pipeline includes an end gate valve and a side discharge valve connected by a T-junction. During the experiments, the end gate valve remains closed. The flow through the side discharge valve was dynamically controlled by a voice coil actuator, allowing precise manipulation of flow conditions to generate transient events. The pipe is equipped with five pressure sensors (P1 to P5) distributed along its length, which allow for detailed monitoring of the transient pressure wave behavior. The sampling rate of the pressure sensors is 2000 Hz.

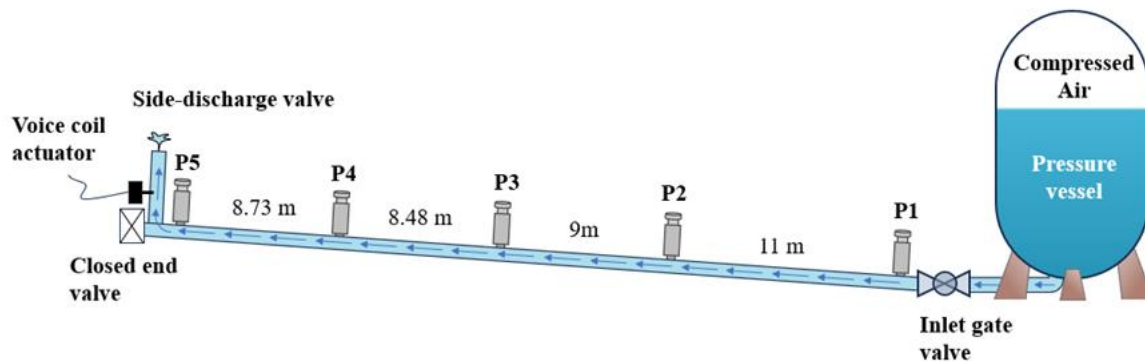


Fig. 14 A schematic of the laboratory pipeline system.

To generate the real dataset, 100 random closing curves were created as input signals to control the voice coil actuator. The initial air pressure in the pressure vessel was set to 200 kPa. The initial flow through the side discharge valve was 0.11 L/s. A period of 0.4 s, in which the

pressure wave can propagate back and forth a couple of times, was selected for the analysis. The real dataset is split into training, validation, and test sets with proportions of [0.8, 0.1, 0.1]. For the simulated dataset, pressure heads measured at P1 and P5 in the real dataset for training were used as boundary conditions. Then, the MOC model with a time step of 0.0005 s was performed to generate the simulated dataset.

Fig. 15 shows the measured and simulated data for ten transient scenarios. Small discrepancies are observed between the measured and simulated results, which can be attributed to unmodeled damping effects, inaccurate system parameters, and measurement errors. Despite these discrepancies, the simulated dataset provides sufficient data across the spatiotemporal domain, offering a physics-consistent representation of the system that complement the physics-driven component in the hybrid loss. The temporal resolution of both the real dataset and simulated dataset is downsampled to 500 Hz and the spatial resolution of the simulated dataset is downsampled to 36 evenly-distributed locations along the pipeline. Due to the insensitivity of the weight w_f to the FPINO performance, the weights values are set as $w_f = 0$, $w_{sim} = 1$, and $w_{real} = 1$.

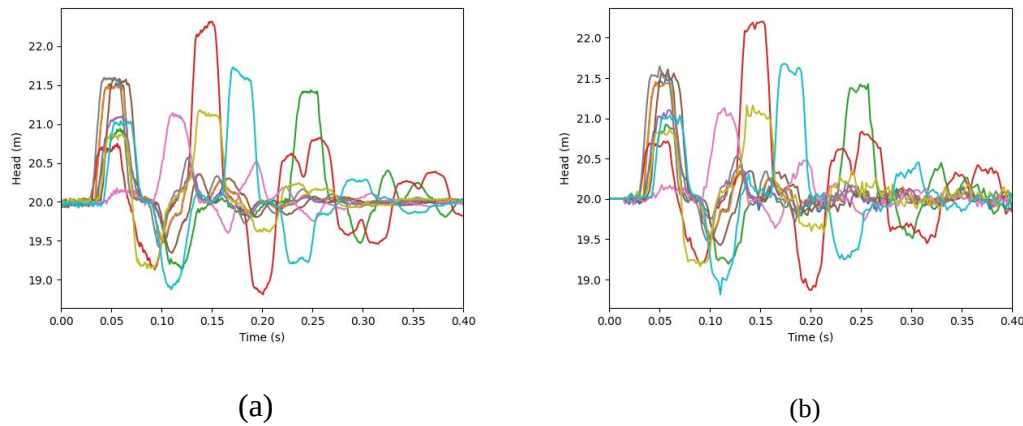


Fig. 15 Ten samples of pressure at P3: (a) Measured (b) Simulated.

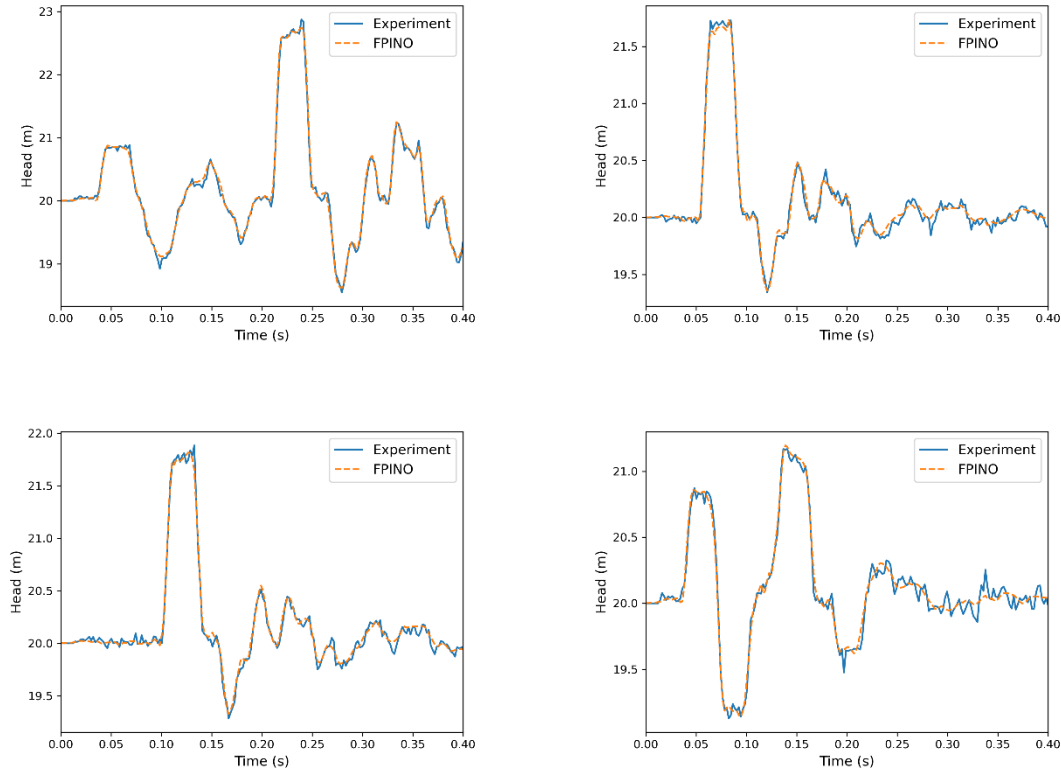


Fig. 16 Comparison of predicted and experimental pressure heads at P4 in four transient scenarios.

Fig. 16 compares the experimental results with the predictions of the FPINO model for transient pressure head measurements at P4 across four scenarios in the test dataset. The FPINO model captures the transient behavior effectively, although slight discrepancies are observed in the amplitude of some smaller peaks. These differences may be attributed to noise in the experimental data and challenges in capturing high-frequency oscillations in specific regions. Overall, the FPINO model provides accurate and robust transient pressure reconstructions across various scenarios. While minor discrepancies exist, these can likely be mitigated through additional tuning of the model's architecture or loss function. The results validate the effectiveness of FPINO for transient reconstruction tasks in hydraulic pipe systems.

5. Conclusions

This study presents a novel FPINO method for near real-time reconstruction of hydraulic transients in pressurized pipelines using sparse sensor data. The proposed method integrates physical constraints through a hybrid loss function and employs a time-weighted information filtering strategy to prioritize causally relevant observations. This improves learning efficiency and prediction accuracy compared to conventional PINO.

FPINO was validated through both numerical simulations and laboratory experiments on single-pipe systems. Results demonstrate that FPINO can effectively reconstruct transient flow behavior under sparse observations and model uncertainties, achieving faster convergence. Balancing the loss weights of physical knowledge, simulation data, and real data is critical for the overall performance and robustness of the model—too much reliance on any can degrade accuracy or generalizability.

However, this work is limited to single-pipe scenarios. The extension of FPINO to more complex configurations, such as water distribution networks, involves additional challenges, including network topology, junction dynamics, and sparse sensor coverage. These aspects will be explored in future work to assess the scalability and adaptability of FPINO in broader engineering applications.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to improve language. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Appendix

Table A1: Summary of L2 errors and MAEs over testing samples of each case.

Case	L2 error (m)		MAE (m)	
	PINO	FPINO	PINO	FPINO
0/400	3.93	3.17	2.29	2.21
50/350	2.92	1.73	1.93	1.22
100/300	2.39	1.84	1.54	1.28
200/200	2.42	2.12	1.53	1.52
300/100	2.78	2.14	1.86	1.75
350/50	2.45	2.15	1.61	1.59
400/0	3.71	2.79	2.46	2.10

Data availability

The data and code are available at: <https://github.com/EthanYe/FPINO.git>

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